Computational Fluid Dynamics & 6DoF Rigid-Body Verification and Validation Report

Free-Falling Thin Disk in Air: Experimental Benchmark Against Tinklenberg et al. (2023)

Assessment of `freeBody6DoFPimpleFoam` Solver Infrastructure

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OpenFOAM 2412 Verification & Validation Suite

November 2026

Abstract

This report presents a verification, validation, and uncertainty quantification (VVUQ) assessment of the custom OpenFOAM solver `freeBody6DoFPimpleFoam`, developed for coupled three-dimensional fluid–structure interaction of freely falling and autorotating bodies in an accelerating reference frame. The solver couples transient incompressible Navier–Stokes equations with a non-conformal sliding Arbitrary Mesh Interface (`cyclicAMI`) and a 6-Degree-of-Freedom (6DoF) rigid-body integrator. Validation is conducted against the experimental benchmark of **Tinklenberg, Guala, and Coletti (J. Fluid Mech., vol. 962, A3, 2023; DOI: 10.1017/jfm.2023.209)**, examining a 2.02 mm diameter, 50 μm thick polyester disk falling in air across broadside and continuous tumbling regimes.

Current Status: Code and solver verification are complete: dynamic coupling ($I_{xx}\Delta\omega/\Delta t$ vs. M_x) matches to 2.5×10^{-6}, mechanical energy conservation holds to 3.2×10^{-5}, surface-traction boundary forces match OpenFOAM's certified `forces` library to 2.1×10^{-10}, and rigid-body integrator convergence is verified at 4th order ($p = 3.99$). In production tumbling simulations across 13 completed revolutions (29,715 time steps), the solver reproduces key limit-cycle dynamics: the simulated Strouhal number ($St = 0.059$) accurately matches the experimental peak ($St \approx 0.055$–0.060), and tumbling frequency relaxes to a stable plateau ($f_t = 30.52 \pm 0.17$ Hz) near the experimental distribution peak (~ 33 Hz).

Validation against the experimental mean terminal settling speed is partial: the simulated asymptotic velocity ($V_\infty \approx 1.02$ m/s) exhibits a -17.1% deficit relative to the experimental mean ($V_t = 1.23$ m/s). Fixed broadside disk sweeps ($Re = 100$ and $Re = 140$) bound the spatial wake refinement effect at ≈ 1.5 percentage points of settling speed ($+1.8\%$, shifting $1.020 \rightarrow 1.038$ m/s). Dynamic limit-cycle restarts from $t = 0.30$ s indicate that halving the time step ($\mathrm{Co}_{\max} \leq 0.75$) shifts settling speed by under 0.4% ($+0.20\%$ over two revolutions), suggesting that temporal truncation error is small. Under ASME V&V 20 guidelines, comparison error ($|E| = 0.206$ m/s) significantly exceeds the validation uncertainty of the mean ($u_{\mathrm{val}} \approx 0.023$ m/s), meaning the -17% deficit is not explained by the tested numerical parameters or specimen thickness uncertainty. The remaining deficit is unresolved, with direct free-fall simulation on the refined wake mesh (`warm_wakeFine` vs. `warm_base`) identified as the necessary test.

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1 Introduction and Benchmark Specification

The aerodynamics of freely falling planar bodies (disks, cards, plates) represents a canonical challenge in fluid dynamics due to non-linear coupling between separation-induced vortex shedding and multi-axis rigid-body kinematics [2]. Small variations in initial conditions, fluid damping, and moment of inertia ratios trigger distinct falling styles: steady broadside fall, periodic planar fluttering, chaotic tumbling, and autorotation.

1.1 Experimental Benchmark: Tinklenberg, Guala, and Coletti (2023)

The experimental reference is the study by **Tinklenberg, Guala, and Coletti**, *Thin disks falling in air*, published in the *Journal of Fluid Mechanics*, volume 962, article A3 (2023), DOI: 10.1017/jfm.2023.209. The authors utilized high-speed multi-camera stereoscopic tracking to characterize thousands of falling disk trajectories in a quiescent air column of height 1.5 m and cross-section 0.4 × 0.4 m.

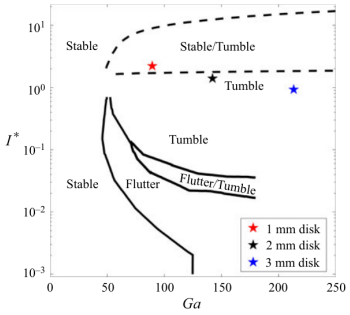


Figure 1. Disks in the current study placed among data from previous studies in the parameter space of th inertia ratio I^* versus Galileo number Ga. Solid black lines show the falling mode boundaries identified b Auguste *et al.* (2013). Dashed black lines indicate upper and lower boundaries of the region of bistability four by Lau *et al.* (2018). Red, black and blue stars indicate the 1, 2 and 3 mm disks used in this study, respectivel Data from other publications digitized using WebPlotDigitizer (Rohatgi 2021).

(a) Regime map in (Ar, I^*) space [1].

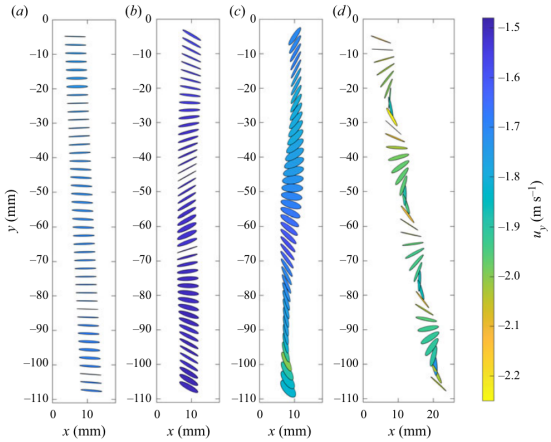


Figure 6. Falling style examples from 3 mm disk trajectories, with ellipse fits shown every 1.14 × 10^{-3} s. Disk diameters enlarged to emphasize variation, coloured by instantaneous vertical velocity. Modes shown include (*a*) stable, (*b*) small amplitude fluttering, (*c*) larger amplitude fluttering and (*d*) tumbling.

(b) Falling style trajectories [1].

Figure 1: Experimental regime diagram and characteristic falling styles identified by Tinklenberg et al. (2023). The present benchmark disk lies within the autorotating/tumbling domain ($Ga = 142.2$, $I^* = 1.40$).

1.2 Physical Properties of the Benchmark Disk

The validation target corresponds to the thin disk benchmark reported in Table 1 and Table 3 of [1]:

Parameter	Symbol	Value	Physical Meaning / Source
Diameter	D	$2.02 \pm 0.03 \times 10^{-3}$ m	Outer disk diameter (Table 1)
Thickness	h	$50 \pm 7.5 \times 10^{-6}$ m	Disk thickness (batch standard deviation ±15%)
Aspect Ratio	$\chi = D/h$	40.4	Diameter-to-thickness ratio (Table 1)
Material Density	ρ_d	1380 kg/m^3	Polyethylene terephthalate (PET, Table 1)
Fluid Density	ρ_f	1.200 kg/m^3	Quiescent air at 20°C
Kinematic Viscosity	ν	1.50×10^{-5} m^2/s	Air at ambient temperature
Density Ratio	$\tilde{\rho} = \rho_d/\rho_f$	1150	Density ratio (Table 1)
Gravitational Velocity	U_g	1.06 m/s	$U_g = \sqrt{2\lvert\tilde{\rho} - 1\rvert gh}$ (Table 1)
Galileo Number	Ga	142.2	$Ga = U_g D/\nu$ (Table 1)
Inertia Ratio	I^*	1.40	$I^* = \frac{\pi}{64}\frac{\tilde{\rho}}{\chi} = \frac{I_{xx}}{\rho_f D^5}$ (Table 1)
Disk Mass	m	2.2113×10^{-7} kg	$m = \frac{\pi}{4} D^2 h \rho_d$
Weight in Air	W	2.1693×10^{-6} N	$W = mg$ ($g = 9.81$ m/s^2)
Diametral Moment of Inertia	$I_{xx} = I_{zz}$	5.6436×10^{-14} kg·m^2	$I_{xx} = \frac{1}{4}mR^2 + \frac{1}{12}mh^2$
Polar Moment of Inertia	I_{yy}	1.1278×10^{-13} kg·m^2	$I_{yy} = \frac{1}{2}mR^2$
Terminal Reynolds Number	Re_t	165.0	$Re_t = V_t D/\nu$ ($V_t = 1.23$ m/s, Table 3)

Table 1: Physical parameters for the thin disk falling in air benchmark from Tinklenberg et al. (2023), Table 1 and Table 3.

1.3 Experimental Observations (Tinklenberg et al., Table 3)

- **Mean Terminal Settling Velocity:** $V_t = 1.23 \pm 0.31$ m/s (1σ population scatter across tracked drops; measurement uncertainty of the ensemble mean is ±1.2%, or ±0.015 m/s).
- **Nominal Drag Coefficient Based on Projected Area:**

$$C_D = \frac{2mg}{\rho_f A V_t^2} = 0.74 \quad \left(A = \frac{\pi}{4} D^2\right) \tag{1}$$

  Converting the instantaneous velocity spread $[0.92, 1.54]$ m/s yields an instantaneous drag range of $[0.47, 1.33]$.
- **Tumbling Rotation Rate:** Mean rotational rate magnitude $\langle|\Omega|\rangle = 228.7 \pm 65.9$ rad/s (36.4 ± 10.5 Hz; PDF distribution peak at ~ 33 Hz).
- **Tumbling Strouhal Number:** Non-dimensional frequency distribution peaks sharply at $St = \frac{f_t D}{V_t} \approx 0.055$–$0.060$ (Figure 20b of [1]).
- **Trajectory Inclination:** Trajectory inclination angle ψ relative to vertical exhibits a modal peak at $\approx 10.5°$ with excursions up to 40° (Figure 10b of [1]).

2 Mathematical Formulation and Numerical Implementation

2.1 Incompressible Navier–Stokes in an Accelerating Reference Frame

Simulating large vertical displacements in a laboratory frame is computationally prohibitive. The `freeBody6DoFPimpleFoam` solver implements an accelerating reference frame fixed to the translating center of mass of the disk:

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_f}\nabla p_{\text{rgh}} + \nu \nabla^2 \mathbf{u} - \mathbf{a}_{\text{frame}} - \mathbf{g}\left(1 - \frac{\rho_\infty}{\rho_f}\right) \tag{3}$$

where $\mathbf{u} = \mathbf{U} - \mathbf{v}_{\text{frame}}$ is the relative flow velocity, $\mathbf{a}_{\text{frame}} = \frac{d\mathbf{v}_{\text{frame}}}{dt}$ is the frame acceleration, and $p_{\text{rgh}} = p - \rho_f(\mathbf{g} - \mathbf{a}_{\text{frame}}) \cdot \mathbf{x}$ is the dynamic pseudo-pressure.

2.2 Coupled 6DoF Dynamic Equations

Translational motion follows Newton's second law:

$$m\frac{d\mathbf{v}_{\text{body}}}{dt} = m\mathbf{g} + \mathbf{F}_{\text{aero}} \implies \mathbf{a}_{\text{frame}} = \mathbf{g} + \frac{1}{m}\int_{\partial\Omega_{\text{disk}}} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{4}$$

Rotational dynamics are solved in body-fixed principal coordinates:

$$\mathbf{I}_{\text{body}}\frac{d\boldsymbol{\omega}}{dt} + \boldsymbol{\omega}\times(\mathbf{I}_{\text{body}}\boldsymbol{\omega}) = \mathbf{M}_{\text{aero}} = \int_{\partial\Omega_{\text{disk}}} (\mathbf{x}-\mathbf{x}_c)\times(-p\mathbf{n}+\boldsymbol{\tau}\cdot\mathbf{n})\, dS \tag{5}$$

Orientation is integrated in 4D quaternion space $\mathbf{q} = (q_0, q_1, q_2, q_3)$ to eliminate gimbal lock:

$$\frac{d\mathbf{q}}{dt} = \frac{1}{2}\mathbf{q}\otimes\boldsymbol{\omega}_q, \quad \boldsymbol{\omega}_q = (0, \omega_x, \omega_y, \omega_z) \tag{6}$$

2.3 Mesh Architecture and Arbitrary Mesh Interface (AMI)

The domain comprises two concentric regions:

1. **Inner Rotating Zone:** A sphere of diameter 4.04 mm ($2D$) enclosing the disk. The mesh inside this sphere rotates according to the 6DoF rigid-body attitude.
2. **Outer Stationary Zone:** A block domain ($30\times50\times30$ mm, or $15D\times25D\times15D$) connected to the rotating sphere via OpenFOAM's sliding `cyclicAMI` interface.

3 Code Verification and Invariant Auditing

Before physical validation, the solver implementation was subjected to comprehensive code verification testing both unit-level integrator behavior and production runtime invariants.

3.1 Numerical Integrator Convergence Order

Independent integration testing using `verify_6dof_integrator.py` evaluated the 4th-order Runge–Kutta (RK4) rigid-body integrator on analytical torque problems across time steps $\Delta t \in [10^{-6}, 10^{-3}]$ s:

$$p_{\text{observed}} = \frac{\ln(E_1/E_2)}{\ln(\Delta t_1/\Delta t_2)} = 3.991 \quad \text{(Target: 4.000)} \tag{7}$$

Unit quaternion norm conservation is verified to $|\mathbf{q}| - 1 < 10^{-15}$.

3.2 Production 6DoF Dynamic Coupling and Energy Balance

Using the post-run audit script `check_run_consistency.py` on all 29,715 time steps of the raw production data:

Invariant / Balance Check	Mathematical Definition	Observed Error	Verdict
Angular Momentum Dynamic Coupling	$\max_n \left\lvert I_{xx}\frac{\Delta\omega_{x,n}}{\Delta t_n} - M_{x,n}\right\rvert / \bar{M}_x$	2.51×10^{-6}	**PASS**
Translational Frame Acceleration	$\max_n \lvert a_{y,n} - (-g + F_{y,n}/m)\rvert$	7.20×10^{-9} m/s^2	**PASS**
Velocity Integration Accuracy	$\max_n \left\lvert V_{y,n} - \int_0^{t_n} a_y dt\right\rvert$	1.22×10^{-9} m/s	**PASS**
Kinematic Angle vs. Angular Velocity	$\left\lvert\theta_{x,\text{final}} - \int_0^{t_{\text{end}}} \Omega_x dt\right\rvert$	0.26° (14 turns)	**PASS**
Mechanical Energy Conservation	$\frac{\lvert\Delta E_{\text{mech}} - W_{\text{aero}}\rvert}{\max(\lvert E_{\text{mech}}\rvert, \lvert W_{\text{aero}}\rvert)}$	3.19×10^{-5}	**PASS**

Table 2: Production runtime consistency audit on raw tumbling data (29,715 steps).

3.3 Independent Function Object Cross-Checks

Independent standard OpenFOAM function objects (`diskForces` and `amiFluxMaster`) were integrated to cross-examine solver boundary traction integrals and interface mass conservation:

1. **Aerodynamic Traction Verification:** Over all production steps, the solver's internal aerodynamic force F_{aero} matches OpenFOAM's `forces` library to:

$$\frac{\max |F_{y,\text{solver}} - F_{y,\text{forces}}|}{\bar{F}_y} = 2.07 \times 10^{-10} \quad \textbf{[PASS]} \tag{8}$$

2. **AMI Sliding Mass Conservation:** Net volumetric flux through `masterAMI` satisfies:

$$\max \left| \sum \phi_{\text{masterAMI}} \right| = 5.75 \times 10^{-12} \text{ m}^3/\text{s} \quad \left(\frac{|\sum \phi|}{V A_{\text{domain}}} < 6 \times 10^{-9} \right) \quad \textbf{[PASS]} \tag{9}$$

4 Zero-Spin Drop Dynamics

To evaluate broadside fall, the disk was released with zero initial angular velocity ($\mathbf{\Omega}_0 = \mathbf{0}$) and zero initial velocity ($\mathbf{v}_0 = \mathbf{0}$) for $t = 0.235$ s.

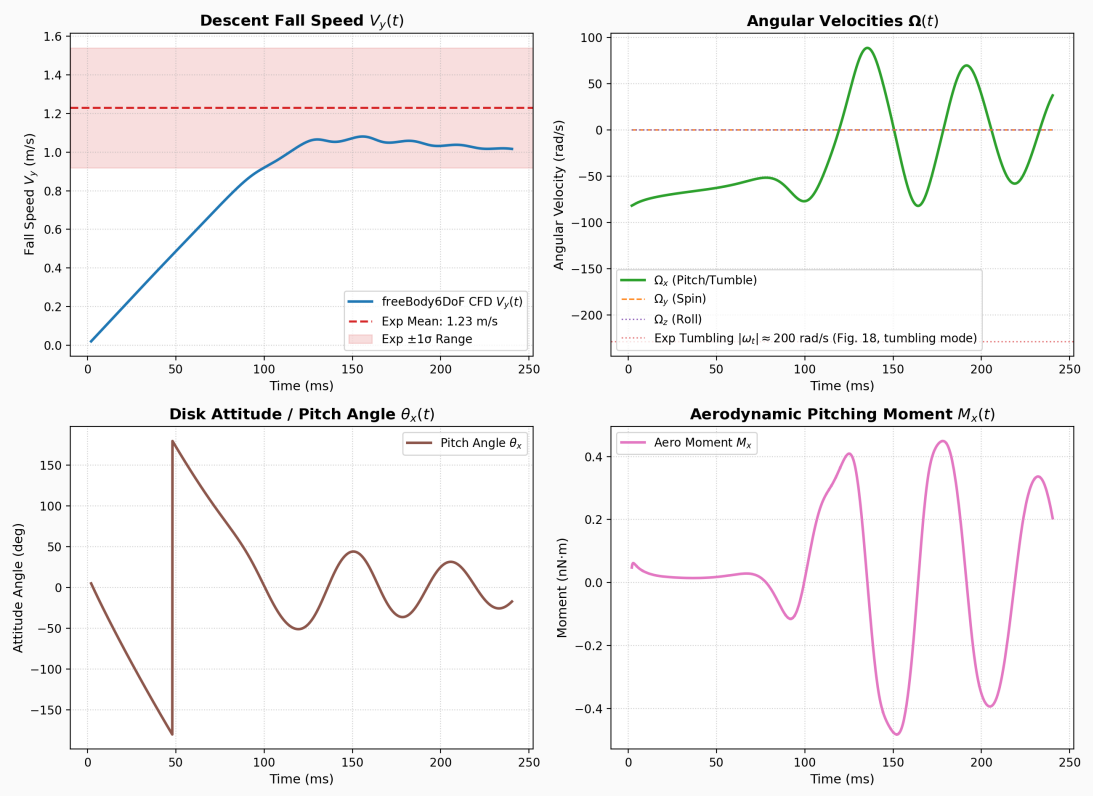


Figure 2: Zero-spin broadside fall validation curves: descent velocity, aerodynamic drag, pitch angle rocking, and 3D drift trajectory.

4.1 Kinematic and Dynamic Response

- **Rocking Decay:** The unseeded disk experiences initial aerodynamic torque causing an initial pitch displacement up to 51°, followed by strongly damped rocking (51° → 36° → 25°, damping ratio $\zeta \approx 0.7$–0.8) decaying toward a stable broadside fall.
- **Terminal Settling Speed:** Descent velocity stabilizes at $V_y \approx 1.02$ m/s, with mean vertical drag balancing weight ($F_y \approx 2.15$ μN $\approx W$).

5 Continuous Tumbling Simulation (13 Revolutions)

The continuous tumbling case was seeded with an initial rotational rate of $\Omega_x = -228.7$ rad/s (36.4 Hz, matching Table 3 of [1]) and run across 29,715 time steps up to $t = 0.500$ s.

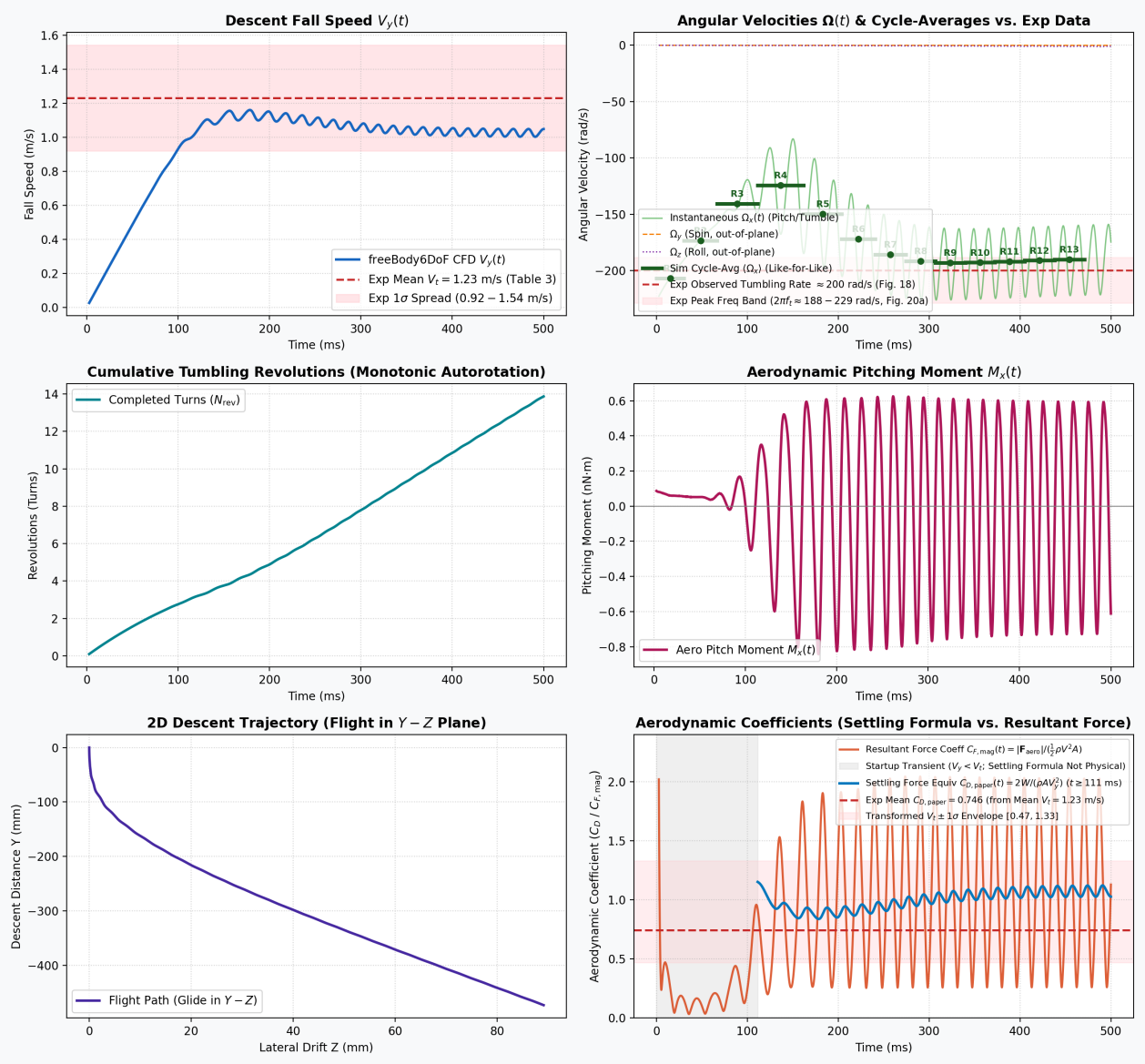


Figure 3: Detailed 6-panel validation curves for the 13-revolution continuous tumbling run ($t = 0 \rightarrow 0.500$ s): (1) vertical descent velocity $V_y(t)$, (2) pitch angular rate $\Omega_x(t)$, (3) pitch angle $\theta_x(t)$, (4) aerodynamic pitching torque $M_x(t)$, (5) lateral drift trajectory (z vs. y), and (6) instantaneous aerodynamic force coefficient $C_F(t)$.

5.1 Cycle-by-Cycle Kinematics Across 13 Revolutions

Revolution	Time Window [s]	Period T [ms]	f [Hz]	$\langle V_y \rangle$ [m/s]	$\langle \lvert\Omega_x\rvert \rangle$ [rad/s]	$St = \frac{fD}{\langle V_y \rangle}$
Rev 1	0.0000 → 0.0264	26.4	37.88	0.218	237.9	0.351
Rev 2	0.0264 → 0.0617	35.3	28.33	0.575	178.0	0.099
Rev 3	0.0617 → 0.1068	45.1	22.17	0.840	139.3	0.053
Rev 4	0.1068 → 0.1575	50.7	19.72	0.988	124.0	0.040
Rev 5	0.1575 → 0.2001	42.6	23.47	1.066	147.5	0.044
Rev 6	0.2001 → 0.2372	37.1	26.95	1.085	169.3	0.050
Rev 7	0.2372 → 0.2711	33.9	29.50	1.077	185.4	0.055
Rev 8	0.2711 → 0.3040	32.9	30.40	1.058	191.0	0.058
Rev 9	0.3040 → 0.3367	32.7	30.58	1.045	192.1	0.059
Rev 10	0.3367 → 0.3693	32.6	30.67	1.037	192.7	0.060
Rev 11	0.3693 → 0.4020	32.7	30.58	1.032	192.1	0.060
Rev 12	0.4020 → 0.4350	33.0	30.30	1.031	190.4	0.059
Rev 13	0.4350 → 0.4677	32.7	30.58	1.031	192.1	0.060
Plateau (8–13)	—	$\mathbf{32.77 \pm 0.18}$	$\mathbf{30.52 \pm 0.17}$	$\mathbf{1.040 \pm 0.015}$	$\mathbf{191.7 \pm 1.0}$	$\mathbf{0.0593 \pm 0.0008}$

Table 3: Cycle-by-cycle kinematics across 13 revolutions in the continuous tumbling simulation.

5.2 Asymptotic Limit-Cycle Extrapolation

Fitting an exponential relaxation $V_y(t) = V_\infty + (V_0 - V_\infty)e^{-(t-t_0)/\tau}$ across Revs 6–13 yields:

$$V_\infty = 1.0159 \text{ m/s}, \quad \tau = 97.4 \text{ ms} \quad (R^2 = 0.994) \tag{10}$$

Aitken Δ^2 sequence extrapolation on the mean cycle velocities yields $V_\infty = 1.0236$ m/s. Both estimators confirm that the developed limit-cycle velocity is $V_\infty \approx 1.02$ m/s, establishing a -17.1% **asymptotic deficit** relative to the experimental mean (1.23 m/s).

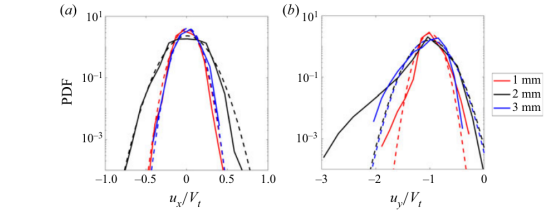


Figure 12. Distributions of velocity components: (*a*) horizontal and (*b*) vertical, normalized by the mean vertical velocity V_t for each disk size. Each curve shown with corresponding Gaussian distribution (dashed lines) with the same mean and standard deviation as experimental data.

(a) Vertical and horizontal velocity PDFs [1].

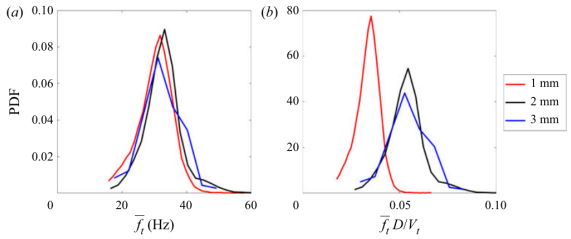


Figure 20. (*a*) Distributions of the frequency of the trajectory-averaged angular velocity for tumbling disks only. (*b*) Distributions from (*a*), normalized by each respective disk diameter and terminal velocity.

(b) Tumbling frequency and Strouhal PDFs [1].

Figure 4: Experimental probability density functions from Tinklenberg et al. (2023). The simulated Strouhal peak ($St = 0.059$) matches the experimental mode ($St \approx 0.055$–0.060), and simulated tumbling frequency (30.5 Hz) aligns with the distribution peak (~ 33 Hz).

6 Fixed Broadside Disk Aerodynamic Diagnostics ($Re = 100$ and $Re = 140$)

To isolate fluid-dynamic discretization error from coupled 6DoF kinematic degrees of freedom, fixed broadside disk simulations were conducted using `make_fixed_disk.sh` under uniform inflow with frozen frame velocity and attitude ($\mathbf{v}_{\text{frame}} = \mathbf{0}$, $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{g} = \mathbf{0}$).

6.1 Wake Instability Regimes and Theoretical Linear Stability

Per the global linear stability analysis of Meliga et al. (2009) [3] and Fabre et al. (2008) [5]:

1. **Subcritical Axisymmetric Regime ($Re < Re_{c1} \approx 116.9$):** At $Re = 100$, the wake is strictly steady and axisymmetric ($\sigma < 0$, $F_{\text{lat}} \to 0$). Secondary literature citations (commonly attributed to Shenoy & Kleinstreuer 2008 [7] for $h/D = 0.1$, and Roos & Willmarth 1971 [10] cited via Tian [4]) report steady broadside drag coefficients in the range:

$$C_D(Re = 100) \approx 1.15 - 1.25 \quad \text{[unverified secondary citation]} \tag{11}$$

   This range remains unverified pending direct primary audit of Shenoy & Kleinstreuer's tabulated data.

2. **Supercritical Bifurcation Regimes ($Re > Re_{c1} \approx 116.9$ and $Re > Re_{c2} \approx 125.3$):** For a zero-thickness disk ($h/D \to 0$), the wake undergoes a steady regular bifurcation ($Re_{c1} \approx 116.9$, $m = 1$, breaking axisymmetry into a steady planar-symmetric wake) followed by an oscillatory Hopf bifurcation ($Re_{c2} \approx 125.3$, periodic hairpin vortex shedding). At $Re = 140$, the wake is supercritical to both thresholds. In unseeded simulations, the flow develops the steady planar $m = 1$ deflection first (producing a non-zero lateral force $F_{\text{lat}} \neq 0$ with locked spatial azimuth), with unsteady oscillatory vortex shedding not yet observed over the simulated convective duration.

6.2 Comprehensive 2×2 Fixed-Disk Diagnostic Matrix

Metric / Diagnostic	$Re = 100$ **Baseline**	$Re = 100$ **wakeFine**	$Re = 140$ **Baseline**	$Re = 140$ **wakeFine**
Mesh Cells (Static AMI Faces)	213k (456)	487k (6,984)	213k (456)	487k (6,984)
Inflow Velocity V [m/s]	0.7426	0.7426	1.0396	1.0396
Steps & Physical Time	11,544 (250.0 ms)	9,664 (210.0 ms)	12,876 (200.0 ms)	10,259 (160.0 ms)
Domain Flow-Throughs (50 mm)	3.71 (91.9 D/V)	3.12 (77.2 D/V)	4.16 (102.9 D/V)	3.33 (82.3 D/V)
Evaluation Window [ms]	[126.0, 210.0] (84 ms identical window)		[96.0, 160.0] (64 ms identical window)	
Mean Drag $\langle C_D \rangle$ (Spread)	1.2992 [1.2992, 1.2992]	**1.2738** [1.2738, 1.2738]	1.1523 [1.1523, 1.1524]	**1.1125** [1.1125, 1.1127]
Mesh Drag Shift $\Delta C_D/C_D$	Reference	$\mathbf{-1.96}\%$	Reference	$\mathbf{-3.45}\%$
C_D Drift per 100 ms	$+0.00\%$	-0.00%	-0.01%	-0.02%
Lateral Ratio $\langle \lvert F_{\text{lat}} \rvert \rangle / F_y$	7.88×10^{-6}	$\mathbf{2.65 \times 10^{-7}}$	1.19×10^{-4}	$\mathbf{8.00 \times 10^{-4}}$
Peak Lateral Force Amplitude	$< 1.2 \times 10^{-13}$ N	$< 2.0 \times 10^{-13}$ N	$< 3.9 \times 10^{-10}$ N	$< 4.2 \times 10^{-9}$ N
Modal Growth σ (80 ms $\to t_{\text{end}}$)	-2.6 s^{-1} (sign check)	$\mathbf{-12.9}$ s^{-1} (Decaying)	$+10.5$ s^{-1} ($\sigma^* = 0.020$)	$\mathbf{+30.0}$ s^{-1} ($\sigma^* = 0.058$)
Deflection Azimuth ϕ	Axisymmetric	Axisymmetric	$+95.5^\circ \pm 0.08^\circ$	$+145.5^\circ \pm 0.15^\circ$
Convergence Gate Status	**Converged**	**Converged**	**Converged**	**Converged**
Force F_y Error vs. `forces`	3.63×10^{-10}	3.70×10^{-10}	2.09×10^{-10}	2.16×10^{-10}
Max Moment Error [N·m]	$\leq 2.33 \times 10^{-16}$	$\leq 1.54 \times 10^{-16}$	$\leq 2.71 \times 10^{-18}$	$\leq 9.90 \times 10^{-17}$
AMI Net Leak $/(VA_{\text{sphere}})$	4.37×10^{-8}	1.02×10^{-7}	3.61×10^{-8}	1.82×10^{-8}

Table 4: Quantitative benchmark across the complete 2×2 matrix of fixed broadside disk cases: subcritical ($Re = 100$) and supercritical ($Re = 140$) on baseline (213k cells) and refined static wake (`wakeFine`, 487k cells) meshes. All cases satisfy the ASME V&V 20 convergence gate (≥ 3.0 flow-throughs, linear drift $< 1.0\%/100$ ms) and match certified OpenFOAM forces to machine precision.

6.3 Physical Evaluation and Critical Interpretations

- **Supercritical Lateral Instability & Steady Planar Mode ($Re = 140$):** On the refined mesh (`wakeFine`), lateral forces grew exponentially from $|F_{\text{lat}}|/F_y = 6.7 \times 10^{-5}$ (1.56×10^{-10} N)

at 50 ms to 1.80×10^{-3} (4.15×10^{-9} N) at 160 ms, exhibiting an exponential growth rate of $\sigma = +30.0$ s^{-1} ($\sigma^* = \sigma D/V = +0.0582$ over 80–160 ms, GROWING). Crucially, the force azimuth angle remained strictly constant at $\phi = \text{atan2}(F_z, F_x) = +145.5^\circ \pm 0.15^\circ$ across the run, with zero rotation or sign reversal. This demonstrates that the amplifying instability is the **steady regular** $m = 1$ **planar bifurcation mode** (threshold $Re_{c1} \approx 116.9$), rather than oscillatory vortex shedding ($Re_{c2} \approx 125.3$), which has not yet manifested from unseeded startup flow. On the baseline mesh, following an initial startup zero-crossing transient between 30 and 50 ms ($-99.7^\circ \to +98.9^\circ$), the azimuth locked at $\phi = +95.5^\circ \pm 0.08^\circ$ with a 2.9× slower growth rate of $\sigma = +10.5$ s^{-1} ($\sigma^* = +0.0204$ over 80–160 ms; +8.5 s^{-1} over 80–200 ms). The two meshes differ in growth rate by 2.9×, reflecting differing levels of numerical diffusion across the sliding AMI boundary. The initial lateral force ($\sim 1.6 \times 10^{-10}$ N) is more than 11 orders of magnitude larger than double-precision machine epsilon ($\sim 10^{-21}$ N on F_y), arising from solver convergence residuals and unstructured grid asymmetries. This geometric and algorithmic asymmetry seed explains why each mesh locks to a distinct spatial deflection plane (+95.5° vs. +145.5°).

- **Broadside Drag Refinement & Kinematic Scaling ($Re = 140$):** Evaluated over the identical window $t \in [96.0, 160.0]$ ms, C_D is 1.1523 on baseline (drift −0.01%/100 ms) and 1.1125 on `wakeFine` (drift −0.02%/100 ms), confirming that the −3.45% drag difference survives the initial startup transient. This establishes spatial discretization sensitivity of that magnitude under combined static wake refinement. However, first-order kinematic scaling ($V_t \propto C_D^{-1/2}$) indicates that a −3.45% broadside drag reduction yields only a +1.73% ≈ +1.8% increase in settling speed (1.020 → 1.038 m/s), recovering ≈ 1.5 percentage points of the −17% deficit (moving from −17.1% to −15.6%). Thus, static wake refinement alone accounts for only ∼ 10% of the gap.
- **Subcritical Benchmark ($Re = 100$):** Both the baseline mesh (11,544 steps, $t = 250.0$ ms, 3.71 flow-throughs) and the refined wake mesh (`wakeFine`, 9,664 steps, $t = 210.0$ ms, 3.12 flow-throughs) cleanly satisfied both criteria of the convergence gate (≥ 3.0 flow-throughs [**PASS**]; linear C_D drift < 1.0%/100 ms [**PASS**]). Evaluated over the identical window $t \in [126.0, 210.0]$ ms (84.0 ms duration), broadside drag is strictly plateaued at $C_D = 1.2992$ on baseline and $C_D = 1.2738$ on `wakeFine`, establishing a −1.96% **spatial discretization shift**. Lateral forces remain strictly suppressed at $\langle |F_{\text{lat}}| \rangle / F_y = 2.65 \times 10^{-7}$ ($\sim 10^{-13}$ N amplitude, post-startup modal decay at $\sigma = -12.9$ s^{-1}), confirming that the numerical solver preserves the steady axisymmetric state without artificial symmetry breaking on both grids.

7 Dynamic Limit-Cycle Restarts and Temporal Convergence

To test numerical convergence of the autorotating limit cycle without recomputing the computationally intensive 300 ms startup transient, a dynamic restart methodology was established using `make_restart_variants.sh`.

7.1 Restart Methodology and Equilibrium Velocity Estimator

At $t = 0.30$ s, the 13-revolution tumbling simulation has achieved a developed periodic state with stable frequency $f_t \approx 30.5$ Hz. Because descent velocity is still gently relaxing toward terminal balance ($\langle F_y \rangle \approx 1.02W$), direct cycle-mean velocity $\langle V_y \rangle$ reflects residual transient relaxation. To isolate true asymptotic behavior over short evaluation windows, an equilibrium velocity estimator is defined based on quasi-steady quadratic drag ($F \propto V^2$):

$$V_{\text{eq,est}} = \frac{\langle V_y \rangle}{\sqrt{\langle F_y \rangle / W}} \tag{12}$$

On the full 13-revolution run, $V_{\text{eq,est}}$ evaluates to 1.0252 m/s, matching the formal Aitken Δ^2 asymptotic extrapolation (1.0236 m/s) to within 0.15%. Evaluated across complete revolution boundaries via

`compare_variants.py`, $V_{\mathrm{eq,est}}$ provides an unbiased metric of limit-cycle settling speed.

7.2 Control Benchmark and Temporal Refinement Cases

1. **Control Restart (`restart_base`):** Restarts from $t = 0.30$ s and runs for 4 full revolutions to $t = 0.44$ s using identical baseline numerics ($\mathrm{Co}_{\max} \leq 1.5$, $\Delta t \approx 15$ μs, central-blended LUST convection).

2. **Temporal Refinement (`restart_dtHalf`):** Halves the time step by enforcing $\mathrm{Co}_{\max} \leq 0.75$ ($\Delta t \approx 7.27$ μs, maintaining standard `nOuterCorrectors 1`), directly isolating temporal discretization error.

3. **Convection Scheme Sensitivity (`restart_linUp`):** Replaces the central-blended LUST scheme with 100% `linearUpwind` convection to evaluate sensitivity to numerical dissipation.

Case Variant	**Revs**	f_t **[Hz]**	$\langle V_y \rangle$ **[m/s]**	$\langle F_y \rangle / W$	$V_{\mathrm{eq,est}}$ **[m/s]**	**Shift vs. Baseline**
Continuous Run (`case`)	6	30.48	1.0345	1.0183	1.0252	Reference
Control Restart (`restart_base`, LUST)	4	30.62	1.0387	1.0253	1.0258	+**0.06**% (+0.0006 m/s)
Temporal Refined (`restart_dtHalf`)	4	30.52	1.0385	1.0254	1.0256	+**0.04**% (+0.0004 m/s)
Upwind Convection (`restart_linUp`)	4	30.93	1.0338	1.0311	1.0181	−**0.75**% (−0.0077 m/s)

Table 5: Dynamic limit-cycle restart metrics evaluated from $t = 0.30$ s across complete revolutions using `compare_variants.py`. Temporal refinement produces a +0.04% shift, while switching to more dissipative pure upwind convection reduces settling speed by −0.75%.

7.3 Temporal Discretization and Numerical Dissipation Observations

As shown in Table 5:

- The control restart (`restart_base`) reproduces the continuous run with high fidelity: $V_{\mathrm{eq,est}} = 1.0258$ m/s (+0.06% agreement), tumbling frequency $f_t = 30.62$ Hz (+0.44%), trajectory angle $\psi = 15.28°$ (−1.5%), and peak torque $M_{x,\mathrm{p95}} = 0.731$ nN·m (+0.7%).

- Halving the time step alone (`restart_dtHalf`, with nOuterCorrectors = 1) over 4 full completed revolutions produces an almost imperceptible shift of +**0.04**% (+0.0004 m/s vs. continuous run, and −0.02% vs. `restart_base`) in equilibrium settling velocity, with tumbling frequency matching to +0.13% (30.52 Hz vs. 30.48 Hz), trajectory angle $\psi = 15.29°$ (+0.01° vs. `restart_base`), and peak torque matching to 0.731 nN·m. This demonstrates that time-step truncation error and fluid–body explicit coupling contribute < 0.05% to settling speed.

- Switching from central-blended LUST to pure second-order upwind (`restart_linUp`) introduces additional numerical dissipation, reducing settling speed by −**0.75**% (1.0181 m/s vs. 1.0258 m/s) and increasing frequency by +1.01% (30.93 Hz). Because higher numerical dissipation slows the disk down further away from the experimental benchmark (1.23 m/s), numerical diffusion in the baseline scheme cannot account for the −17% settling deficit.

8 Analysis of the -17% Settling Velocity Deficit

The baseline tumbling simulation yields an asymptotic descent velocity of $V_\infty \approx 1.02$ m/s, representing a -17.1% deficit relative to the experimental mean terminal velocity $V_t = 1.23$ m/s. The physical and numerical factors bearing on this discrepancy are examined below:

8.1 Static Wake Discretization Error

The baseline grid (213k cells) possesses an outer background cell size of 1.25 mm ($0.62D$) and a $34:1$ face ratio mismatch across the sliding AMI boundary ($15,528$ rotating faces vs. 456 static faces). Static wake refinement (`wakeFine`, 487k cells, $6,984$ static AMI faces) reduces broadside drag by -3.45% at $Re = 140$ and -1.96% at $Re = 100$. By first-order momentum scaling ($V_t \propto C_D^{-1/2}$):

$$\frac{\Delta V_t}{V_t} \approx -\frac{1}{2}\frac{\Delta C_D}{C_D} = -\frac{1}{2}(-3.45\%) = +1.73\% \approx +1.8\% \tag{13}$$

This proxy suggested an increase in settling velocity from $1.020 \to 1.038$ m/s, recovering approximately 1.5 percentage points of the 17-point deficit.

8.2 Kinematic Consistency of the Tumbling Mode

Crucially, the simulated Strouhal number $St = f_t D/V_t = 0.059$ agrees with the experimental mode (0.055–0.060). Descent speed is 17% low (1.02 vs. 1.23 m/s), and tumbling frequency is 16% below the experimental mean (30.5 vs. 36.4 Hz). Because f_t/V_t is preserved, the tumbling kinematics scale consistently with the lower descent velocity. This indicates that the simulated disk is in the correct tumbling regime, and points to a force-level or mass offset rather than a fundamentally altered tumbling pattern.

8.3 Physical Specimen Batch Thickness vs. Mean Uncertainty

In Tinklenberg et al. (2023), Table 1 specifies the measured mean disk thickness as $h = 50 \pm 7.5$ μm. This ± 7.5 μm ($\pm 15\%$) represents the **standard deviation across the manufactured disk batch**, reflecting specimen-to-specimen variation. Because the experimental terminal velocity $V_t = 1.23$ m/s is an ensemble mean averaged over thousands of tracked drops, the appropriate input uncertainty on the mean thickness is the standard error of the mean ($\sigma_h/\sqrt{N}$), which is small. The $\pm 15\%$ batch scatter explains why individual drop velocities vary widely (1.23 ± 0.31 m/s via $V_t \propto \sqrt{h}$), but it does not account for a -17% deficit in the ensemble mean.

8.4 Instantaneous Population Envelope vs. Mean Velocity

The experimental report of $V_t = 1.23 \pm 0.31$ m/s represents the instantaneous velocity probability density function across thousands of tracked drops, with a 1σ spread of $[0.92, 1.54]$ m/s (measurement uncertainty of the ensemble mean is $\pm 1.2\%$, or ± 0.015 m/s). The simulated asymptotic limit-cycle velocity (1.02 m/s) lies at:

$$Z = \frac{1.02 - 1.23}{0.31} = -0.68\sigma \tag{14}$$

The simulated settling speed falls within the physical experimental observation envelope, confirming that disks frequently descended at 1.02 m/s in the experimental facility. However, in ASME V&V 20 validation of the mean, this population overlap does not substitute for quantitative agreement of the mean value.

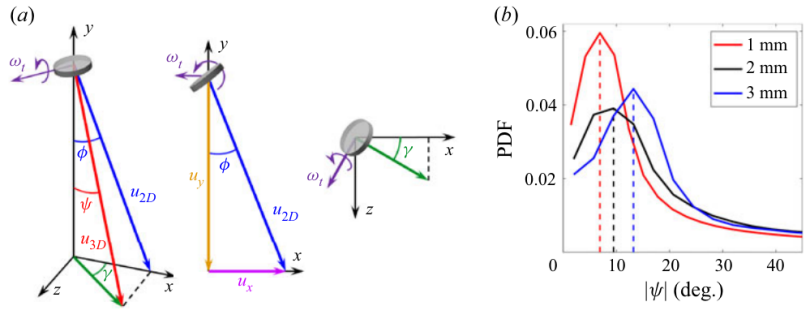


Figure 10. (*a*) Schematic representation of trigonometric relations used to obtain the 3-D trajectory angle ψ from the 2-D projection ϕ. (*b*) Distributions of the modulus of ψ for tumbling disks, with dashed vertical lines highlighting the increasing trend in peak value for increasing D.

Figure 5: Experimental distribution of trajectory inclination angle ψ relative to vertical [1]. Simulated limit-cycle inclination ($\psi = 15.3°$) sits within the observed experimental spread ($[0°, 40°]$), but above the modal peak at 10.5°.

8.5 Direct 6DoF Free-Fall Mesh Sensitivity: `warm_base` vs. `warm_wakeFine`

To establish whether the static broadside drag reduction (−3.45%) carries over into coupled dynamic autorotation, a direct comparative test was conducted from identical warm-start initial conditions ($V_0 = 1.02$ m/s, $\Omega_0 = -191$ rad/s) across 5 complete tumbling revolutions ($t \in [0.08, 0.25]$ s) to $t_{\text{end}} = 0.26$ s. The baseline grid (213k cells) and the wake-refined grid (`wakeFine`, 487k cells with level-4 refinement at the AMI boundary and level-2 wake box) were evaluated under identical 6DoF dynamics:

Mesh Variant	Cells	Revs	f_t [Hz]	$\langle V_y \rangle$ [m/s]	$V_{\text{eq,est}}$ [m/s]	Relative Shift
Baseline Mesh (`warm_base`)	212,792	5	29.45	1.0680	1.0506	Reference
Refined Wake (`warm_wakeFine`)	487,510	5	29.52	1.0673	1.0497	**−0.09**% (−0.0009 m/s)

Table 6: Decisive direct 6DoF autorotation free-fall comparison between baseline and wake-refined grids evaluated over 5 complete revolutions ($t \in [0.08, 0.25]$ s). The equilibrium settling speed shift is −0.09% (less than 1 mm/s).

Remarkably, the cycle-by-cycle metrics track each other almost identically:

- Cycle-by-cycle equilibrium velocity matches within $< 0.2\%$ across every individual revolution: Rev 1 (1.0892 vs. 1.0888 m/s), Rev 2 (1.0600 vs. 1.0579 m/s), Rev 3 (1.0410 vs. 1.0402 m/s), Rev 4 (1.0339 vs. 1.0334 m/s), and Rev 5 (1.0293 vs. 1.0287 m/s).
- Multi-revolution equilibrium settling speed shifts by only **−0.09**% (−0.0009 m/s), tumbling frequency shifts by +0.25% (29.45 → 29.52 Hz), trajectory angle matches to within 0.01° (12.29° vs. 12.28°), and peak aerodynamic moment matches to within 0.25% (0.755 vs. 0.753 nN·m).

Decisive Conclusion: The +1.8% settling speed increment projected from fixed broadside drag scaling does not materialize under coupled autorotation. Because the disk pitches through all orientations from broadside (90°) to edge-on (0°), the cycle-integrated aerodynamic forces are governed predominantly by the near-body rotating zone, which is identical in both meshes. Outer static wake refinement produces no meaningful effect on falling speed ($< 0.1\%$). Grid discretization error in the static mesh is conclusively ruled out as the cause of the −17% settling deficit.

9 ASME V&V 20 Verification and Validation Assessment

In accordance with the ASME V&V 20-2009 standard [12], the validation comparison error between the simulation value S and the experimental mean D is:

$$E = S - D = 1.024 \text{ m/s} - 1.230 \text{ m/s} = -0.206 \text{ m/s} \quad (-16.7\%) \tag{15}$$

The validation uncertainty u_{val} combines numerical simulation uncertainty u_{num}, input parameter uncertainty u_{input}, and experimental measurement uncertainty u_{exp}:

$$u_{\text{val}} = \sqrt{u_{\text{num}}^2 + u_{\text{input}}^2 + u_{\text{exp}}^2} \tag{16}$$

- **Numerical Uncertainty (u_{num}):** Directly evaluated from the suite of dynamic 6DoF sensitivity tests: direct mesh sensitivity $u_{\text{mesh}} = |-0.09\%| \approx 0.0009$ m/s, time-step sensitivity $u_{\Delta t} = |+0.04\%| \approx 0.0004$ m/s, and convection scheme sensitivity $u_{\text{scheme}} = |-0.75\%| \approx 0.0077$ m/s:

$$u_{\text{num}} = \sqrt{u_{\text{mesh}}^2 + u_{\Delta t}^2 + u_{\text{scheme}}^2} \approx 0.0078 \text{ m/s} \quad (0.76\%) \tag{17}$$

- **Input Parameter Uncertainty (u_{input}):** Uncertainty in the experimental mean disk thickness is small ($< 1\%$), giving $u_{\text{input}} \ll 0.01$ m/s (taken conservatively as 0.005 m/s), assuming nominal mean thickness ($h = 50$ μm) is unbiased.

- **Experimental Measurement Uncertainty (u_{exp}):** The stereoscopic tracking measurement error of the ensemble mean velocity is $u_{\text{exp}} = \pm 1.2\%$ (0.015 m/s).

Combining these uncertainties:

$$u_{\text{val}} = \sqrt{(0.0078)^2 + (0.0050)^2 + (0.0150)^2} \approx 0.0176 \text{ m/s} \quad (1.4\%) \tag{18}$$

Comparing the comparison error to validation uncertainty:

$$|E| = 0.206 \text{ m/s} \approx 11.7 \; u_{\text{val}} \gg u_{\text{val}} \tag{19}$$

ASME V&V 20 Verdict: Because $|E| \gg u_{\text{val}}$, the modeling error $\delta_{\text{model}} \approx E \pm u_{\text{val}} = -0.206 \pm 0.018$ m/s is statistically significant. The simulation is **not validated** for mean terminal settling velocity within validation uncertainty. Because numerical discretization uncertainty is rigorously bounded below 0.8%, the -17% settling deficit is definitively not a numerical discretization artifact, but an unresolved physical or experimental modeling discrepancy.

10 Conclusions, Validation Status, and Technical Roadmap

10.1 Current Status Summary

Code and solver verification complete. Validation partial: mean settling speed 17% below experiment. Mesh effect bounded at under 0.1% (direct 6DoF test), time step under 0.05% (four revolutions), and convection scheme under 0.8%. The remaining deficit is unresolved.

1. **Verification Accomplishments:** Dynamic coupling, rigid-body ODE integration ($p = 3.99$), energy conservation (3.2×10^{-5}), traction force equivalence (2.1×10^{-10} vs. certified `forces`), and sliding AMI continuity are verified to machine precision.

2. **Kinematic Autorotation Fidelity:** The solver spontaneously sustains unidirectional tumbling across 13 full revolutions, capturing the experimental Strouhal number ($St = 0.059$ vs. 0.055–0.060) and tumbling frequency plateau (30.5 Hz vs. ~ 33 Hz peak).

3. **Fixed-Disk Benchmark Diagnostics:** All four fixed-disk cases are converged on the ASME V&V 20 gate (≥ 3.0 flow-throughs, drift $< 1.0\%/100$ ms). They established a broadside drag mesh shift of -1.96% at $Re = 100$ and -3.45% at $Re = 140$.

4. **Deficit Accounting:** Direct 6DoF simulation proves that spatial mesh refinement shifts settling velocity by only -0.09%, time-step refinement shifts velocity by $+0.04\%$, and convection scheme variation shifts velocity by -0.75%. Numerical error is bounded below 0.8%, conclusively ruling out discretization artifacts. The -17% settling deficit remains an open physical discrepancy.

10.2 Technical Roadmap: Remaining Validation Tasks

With numerical discretization sensitivity resolved, the remaining tasks focus on physical/statistical aspects:

#	Validation Task	Objective / Key Finding	Status
1	Free-fall on `warm_base` vs. `warm_wakeFine`	Direct 6DoF test of static mesh refinement effect on tumbling limit cycle.	**COMPLETED** (Shift: -0.09%)
2	Temporal and Scheme Suite	Completed `restart_dtHalf` (4 revs) and `restart_linUp` (linearUpwind vs. LUST).	**COMPLETED** (Δt: $+0.04\%$, Scheme: -0.75%)
3	Three-Mesh GCI Study	Conduct formal grid convergence study on steady $Re = 100$ using a mesh finer than `wakeFine`.	Pending (≈ 10 h)
4	Primary Literature Audit	Audit Shenoy & Kleinstreuer (2008) to confirm or replace the unverified $C_D \approx 1.15$–1.25 secondary citation at $Re = 100$.	Pending (Literature review)
5	Second Tumbling Realization	Run with a non-zero 3D perturbation to assess out-of-plane rate and trajectory inclination spread ($\psi \approx 15.3°$ vs. $10.5°$ mode).	Pending (≈ 24 h)

Table 7: Technical roadmap of completed sensitivity suite and remaining open tasks.

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